Energy Nexus 12 (2023) 100240

Contents lists available at ScienceDirect

Energy Nexus

journal homepage: www.elsevier.com/locate/nexus

An investigation of water gas shift reaction in a Pd-alloy membrane reactor with an optimized crossflow configuration

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ARTICLE INFO

Keywords:
Water gas shift reaction (WGSR)
Palladium (Pd) membrane
Computational fluid dynamics
Fe-Cr catalyst
CO_2 capture
H_2 production and recovery

ABSTRACT

In this study, the performance of a high-temperature water gas shift reaction (WGSR) using a Fe-Cr catalyst along with a Pd alloy membrane was simulated by computational fluid dynamics (CFD). The influences of using Pd membranes, catalytic layer thickness ratio (R/R_0), Reynolds number, and steam-to-CO ratio (S/C) on the reaction were investigated by comparing CO conversion and hydrogen recovery (HR). In the CFD simulation, one-tube and four-tube systems were simulated at 500 °C. This study also compared the performance between tandem and optimized configurations. The results show that the CO conversion can be improved up to 22.9% when the WGSR reactor system uses a Pd membrane compared to the system without a Pd membrane. The system has the best hydrogen recovery performance at S/C = 4 and R/R_0 larger than 1.5. At *Re*=5, the optimized configuration for CO conversion has better performance when R/R_0 is larger than 1.75. Compared to the tandem configuration, the optimized configuration also shows better performance for HR at every R/R_0. The results indicate that a Pd membrane and optimized configuration can significantly improve CO conversion and that R/R_0 and S/C optimization is very important for effective reactor performance.

Introduction

Energy sustainability is one of the pressing global issues today. Accompanying industrial growth, the energy demand of the world is also increasing. A potential energy resource, such as H_2, can be explored to address this problem. H_2 is an important energy carrier or fuel for prospective transport and power systems [1,2]. It can produce electricity in a fuel cell [3] without creating pollutants and greenhouse gases [4]. For this reason, hydrogen utilization, in association with a fuel cell, can also achieve decarbonization [5]. H_2 can be produced from biomass, hydrocarbons, or water [6]. Fuel processing is needed to extract H_2 from these compounds [7].

One of the most important and economical ways of H_2 production is through water gas shift reaction (WGSR) [8]. It is the reaction of carbon monoxide and steam to produce carbon dioxide and water [9]:

$$CO + H_2O \underset{\rightarrow}{\overset{\leftarrow}{}} CO_2 + H_2, \ \Delta H = -40.6\ kJ{\cdot}mol^{-1}$$

It is a reversible and exothermic reaction. Thermodynamically, the gas concentrations in the WGSR at equilibrium depend on the reaction temperature and the feed gas concentrations. The WGSR can be categorized into low-temperature and high-temperature reactions [10,11]. Low-temperature WGSR is operated between 200 and 260 °C using copper, zinc oxide, and alumina as catalysts [12,13]. High-temperature WGSR uses iron and chromium oxides as catalysts between the temperature range of 310 °C and 500 °C [14,15]. The WGSR is an exothermic reaction. Operating at high temperatures is thermodynamically favorable and enables rapid reduction of the CO concentration. However, the CO concentration needs to be further diminished into the ppm scale to meet the standard of the fuel cell application. In that case,

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https://doi.org/10.1016/j.nexus.2023.100240
Received 6 March 2023; Received in revised form 3 September 2023; Accepted 15 September 2023
Available online 16 September 2023
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Nomenclature

C_p	specific heat. ($J{\cdot}mol^{-1}K^{-1}$)
Cp	pressure coefficient (-)
D_{cat}	catalyst diameter (m)
τ	Radial (m)
F_{press}	pressure scale-up factor (-)
H_{WGS}	heat of reaction of WGSR ($J{\cdot}mol^{-1}$)
K_{br}	catalyst bed permeability (m^2)
K_{eq}	equilibrium constant (-)
P	pressure (atm)
q_c	energy source due to the chemical reaction ($J{\cdot}m^{-3}$)
Re	Reynolds number (-)
R	radius of the whole tube (mm)
R_0	radius of the membrane tube (mm)
r_i	reaction rate of species i ($mol{\cdot}m^{-3}{\cdot}s^{-1}$)
T	temperature (K)
$\overline{V}$	Velocity ($m{\cdot}s^{-1}$)
x_i	mole fraction of species *i* (-)

Greek letters

β_F	Forchheimer coefficient ($kg{\cdot}m^{-4}$)
θ	Porosity (-)
μ	Viscosity (Pa·s)
ρ	Density ($kg{\cdot}m^{-3}$)

Subscripts

cat	Catalyst
eff	effective
forward	forward reaction
i	species *i*
in	inlet
j	species *j*
k	species *k*
o	Initial value
out	outlet
p	permeate side
r	retentate side
reverse	reverse reaction
wgs	water gas shift reaction

low-temperature WGSR can help with that, but it is kinetic-oriented, which makes it constrain to catalyst type. The catalysts of high-temperature WGSR have been explored actively in recent years. This study uses a traditional chromium-based hematite catalyst for high-temperature WGSR to achieve rates for industrial applications and resistance against several poisons [16,17].

In H_2 production technologies, fixed-bed reactors have been researched extensively. The performance of reactors can be predicted and improved by simulation. Using computational methods can markedly reduce research costs and duration in complex studies [18]. Gosiewski et al. [19] studied a membrane reactor through mathematical simulation. They addressed that the membrane reactor technology was promising for H_2 production from coal-derived gas under the appropriate temperature. Murgia et al. [20] used a numerical model to compare different catalysts and conditions for WGSR. Their results indicated that the 2-D model was an excellent way to predict WGSR with the commercial catalyst. Palma et al. [21] investigated the influence of carrier thermal conductivity for structured catalysts employing a computational fluid dynamics (CFD) approach. The results showed an intensification of the WGS process, made possible using a structured catalyst with high thermal conductivity.

Reactor design with fluid flow plays a vital role in practical fluid-related reactors, such as membrane reactors and wastewater processors. Adrover et al. [22] used the reactor simulation to compare the different directions of sweep flow in a multitubular membrane reactor. Their results suggested that the inverse (i.e., counter-current) direction was better for H_2 recovery. Alkhamis et al. [23] studied the spacers for gas-gas separation and proposed that the spacers were an essential component of the membrane systems. Jani et al. [24] investigated the structure of helical flow patterns to separate gas in micromixers. They showed that the helical micromixers were favorable for gas-liquid contacting. Chen et al. [25] used evolutionary computation to optimize gas separation. Their results showed that the method could improve the reactor systems efficientl Through the design of the reactor structure and controlled parameters, the performance of reactors can be improved and predicted. This study investigates the crossflow designs of a catalytic membrane reactor (CMR) system. This design is used to deal with more feed gas and tries to increase the CO conversion in WGSR.

To simulate the performances of H_2 permeation and the reaction equilibrium, WGSR is the only reaction in this study. It is known that a membrane can achieve H_2 separation and enhance CO conversion [26]. Through the permeation of H_2, the CMR can overcome the thermodynamic equilibrium limitation to produce more H_2 [27]. For many studies, membranes have been used to separate and recover H_2, as shown in Table 1. However, the sizes of the membrane reactor are limited to the capacity of the reaction gases. The crossflow reactor can contain numerous feed gas and multiple catalyst beds to react. The crossflow design is simulated to analyze the advantages and disadvantages of H_2 production. The influence of Pd membranes is vital for CMR because it helps investigate the enhancement caused by the membrane for WGSR. The CO conversion is compared to one without a membrane to measure the performance of the membrane. For WGSR, the Reynolds number and steam-to-CO molar ratio (S/C ratio) [28] are important factors that influence the reaction. Better performance can also be predicted by changing operating conditions. The optimized configuration of the multiple-tube palladium membrane was investigated previously in a study by Chen et al. [25]. This study applies a multiple-tube palladium membrane optimized configuration to investigate whether it improves CMRs by considering different operating parameters, including catalytic layer thickness ratio (R/R_0), Reynolds number, and steam-to-CO ratio (S/C). The obtained results are conducive to the reactor design and operation to maximize the reactor's performance.

Table 1
. Literature review of different membrane and catalysts for WGSR.

Membrane	Catalyst	Temperature	S/C ratio	CO conversion (%)	H_2 recovery (%)	Ref.
Pd	$Pt/Ce_{0.6}Zr_{0.4}O_2$	250–425 °C	1.5–4	94.9–96.5	48.7–84.8	[49]
SiO_2	Cu-based	220–300 °C	1	95	30	[50]
Pd/Pss	Fe-Cr	390 °C	1–4	80	70	[28]
Pd-Ag	$Pt/La_2O_3{\bullet}SiO_2$	400–450 °C	1–2	98	90	[51]
Pd-Ag	Fe-Cr	350–450 °C	1.1–2.6	98.2	81.2	[45]
Pd/Pss	Fe-Cr	350–450 °C	_	46–66	<15	[52]
Pd/Pss	Fe-Cr	380–450 °C	2.7–3.6	58–85	28–80	[53]

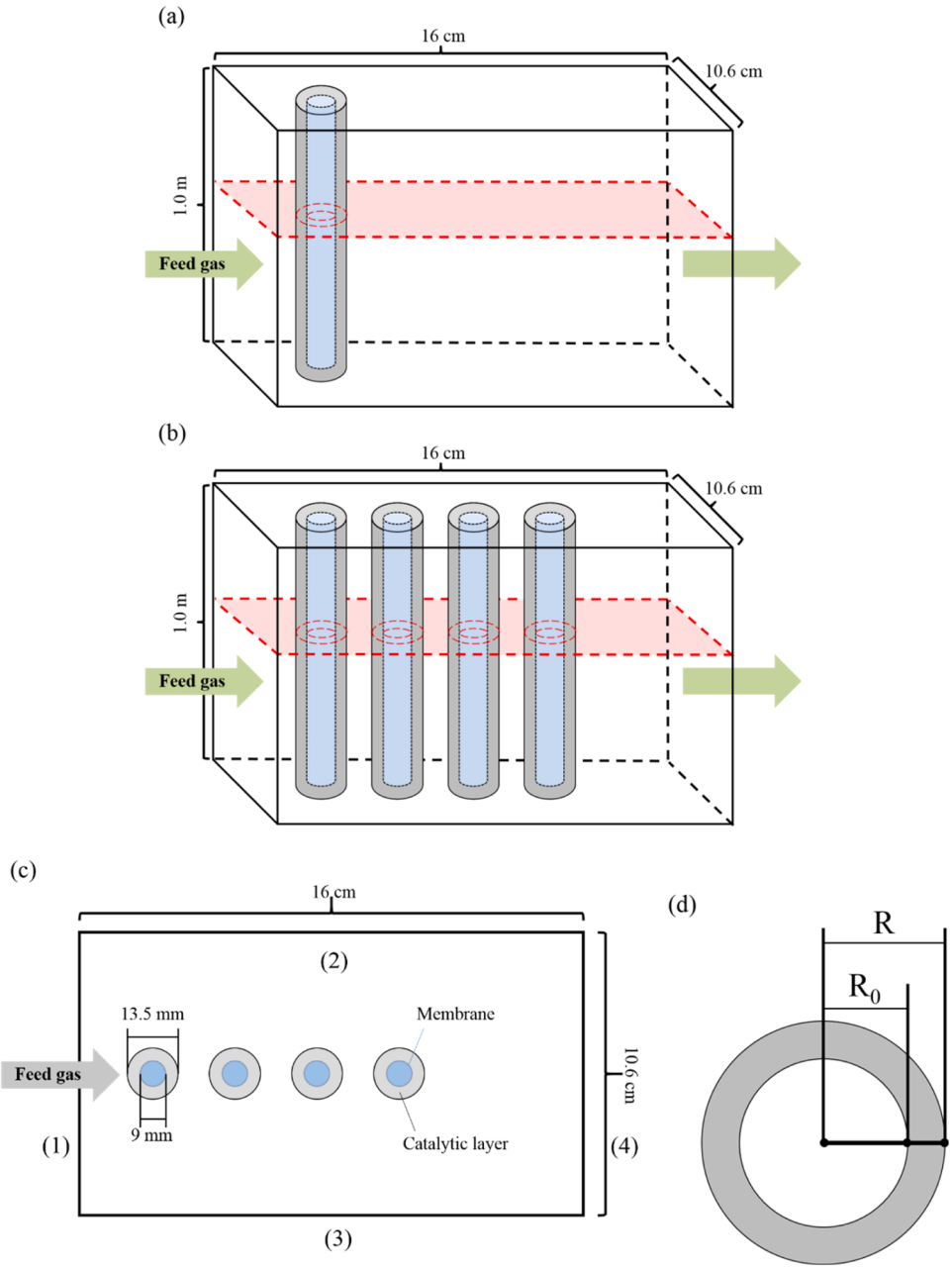


Fig. 1. Schematics of (a) single membrane reactor system, (b) four membrane reactor system geometry, (c) top view, and (d) thickness ratio (R/R_0).

Mathematical formulation

Geometry model

The membrane system used a crossflow structure designed with a single reactor and multi-reactors, shown in Fig. 1**a** and 1**b**. This system included one or more concentric tubes, and the concentric tube consisted of an internal tube and a shell. A Pd-based membrane was the internal tube. The catalyst pellets were placed in the shell, as shown in Fig. 1**c**. The system's length, width, and height were 16 cm, 10.6 cm, and 1 m, respectively. The diameter of the membrane was 9 mm. In this study, the thickness of the catalytic layer was an important parameter to measure the quantity of catalyst. The thickness of the catalytic layer is represented by R/R_0, as shown in Fig. 1**d**. R is the whole tube radius (tube+membrane), and R_0 is the tube radius.

This work adopted several assumptions to simplify the physical problems:

(1) The flow field was laminar in the reactor.
(2) The feed gas was homogeneously mixed.
(3) The H_2 permselectivity of the Pd membrane was infinite.
(4) The catalysts were spherical pellets (diameter = D_{cat}) and the pellets were treated as a homogeneous porous medium with porosity θ and permeability K_{br} [29].

Governing equations and boundary conditions

The governing equations in the non-porous region are the same as those in a previous study [30]. For the porous region, the governing equations are given as follows:

Continuity equation

$$\nabla\cdot\left(\theta\cdot\rho\vec{V}\right)=0 \tag{1}$$

Momentum equations

Table 2
Boundary conditions.

Boundary	Continuity and momentum equations	Energy equation	Species equation
Inlet	$\vec{V} = -V_{in}\vec{n}$	$T = T_{in}$	$w_i = w_{i,in}$
Membrane surfaces	$\vec{V} = 0$	$-k_p\nabla T = -k_r\nabla T$	$\nabla\omega_{CO_2} = \nabla\omega_{CO} = \nabla\omega_{H_2O} = 0$ $-\vec{n}\cdot N = \pm M_{H_2}k(P_{r,H_2}^{0.5} - P_{p,H_2}^{0.5})$
Walls	$\vec{V} = 0$	$\nabla T = 0$	$\nabla w_i = 0$
Outlet	$\nabla\vec{V} = 0$, $P = P_{out}$	$\nabla T = 0$	$\nabla w_i = 0$

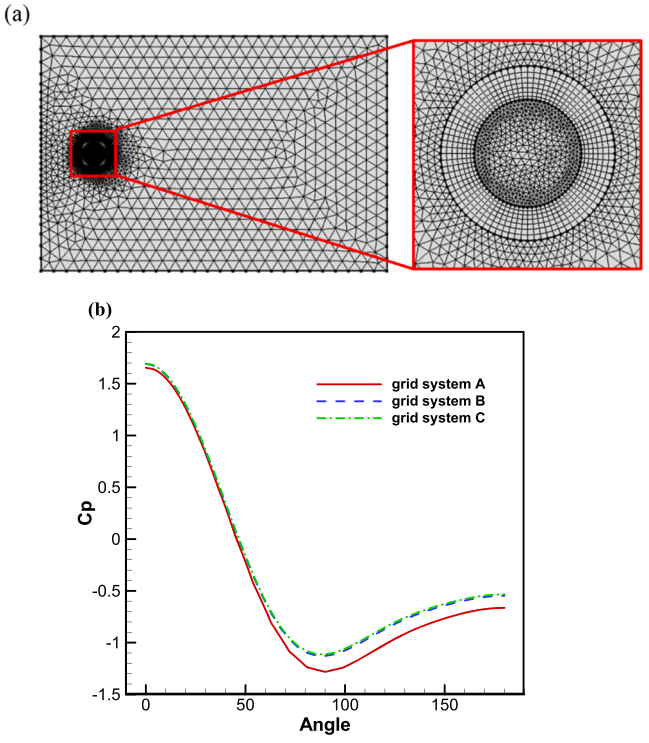


Fig. 2. (a) Grid distributions of the single tube system and (b) pressure coefficient distributions vary with the angle of the membrane tube.

$$\frac{1}{\theta^2}\rho\vec{V}\cdot\nabla\vec{V} = -\nabla p + \nabla\cdot\left[\frac{\mu}{\theta}\left(\nabla\vec{V} + \left(\nabla\vec{V}\right)^T\right) - \frac{2\mu}{3\theta}\left(\nabla\vec{V}\right)\right] + S \tag{2}$$

Species transport equations

$$\nabla\cdot\left(-\rho w_i\sum_j D_{ij,eff}\left(\nabla x_j + \left(x_j - w_j\right)\frac{\nabla p}{p}\right) - D_{ik}^T\left(\frac{\nabla T}{T}\right) + \rho w_i\vec{V}\right) = r_i \tag{3}$$

Equation of state

$$\rho\cdot C_p\vec{V}\cdot\nabla T = \nabla\cdot\left(k_{eff}\nabla T\right) + q_c \tag{4}$$

For the porous momentum equations, the source term, S, is expressed as:

$$S = -\frac{\mu}{K_{br}}\vec{V} + \beta_F\left|\vec{V}\right|\vec{V} \tag{5}$$

where μ is the gas mixture viscosity, K_{br} is permeability, and β_F is the Forchheimer coefficient used for a packed bed with spherical particles. They are written as:

$$K_{br} = \frac{D_{cat}^2\theta^3}{150(1-\theta)^2},\ \beta_F = \frac{1.75\rho}{\sqrt{150K_{br}\theta^3}} \tag{6}$$

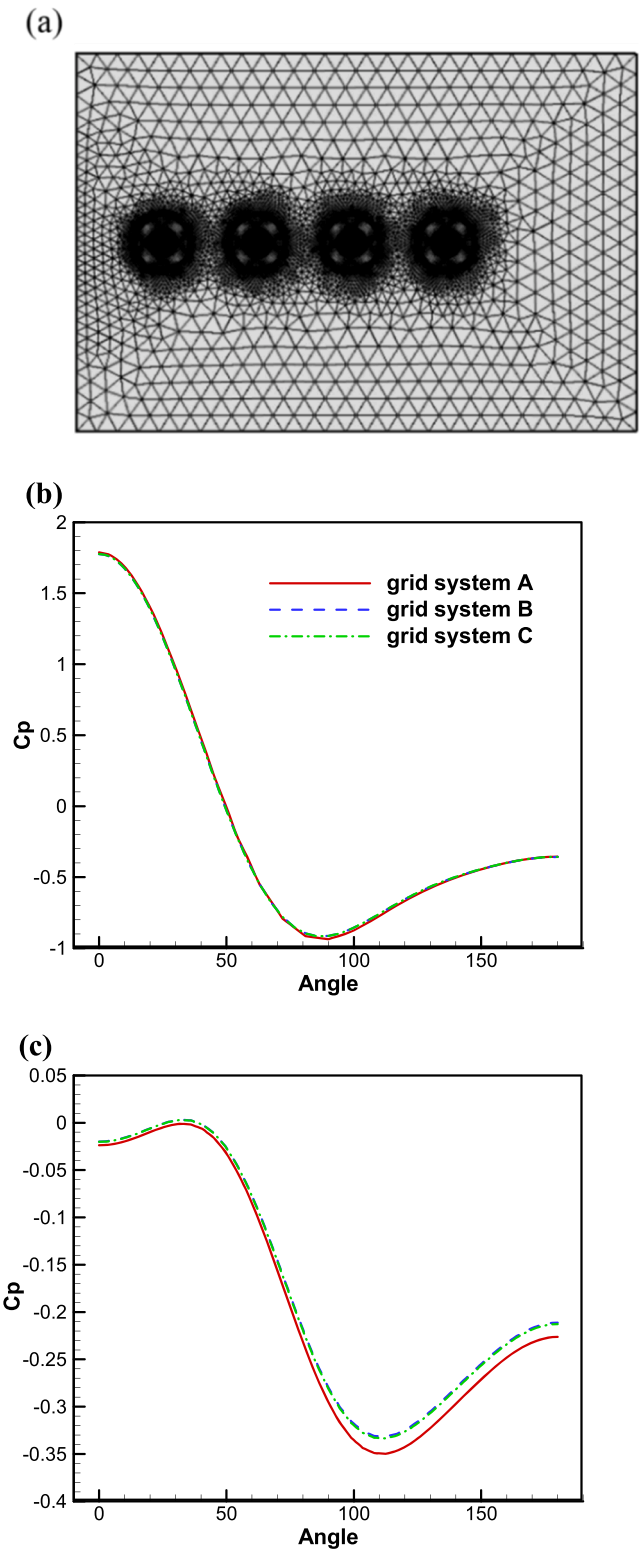


Fig. 3. (a) Grid distributions of four tubes system, pressure coefficient distributions vary with the angle of the membrane tube at (b) the first tube and (c) the fourth tube.

For porous species transport equations. The effective binary gas diffusivity can be approximated as:

$$D_{ij,eff} = D_{ij}\frac{\theta}{\tau} \tag{7}$$

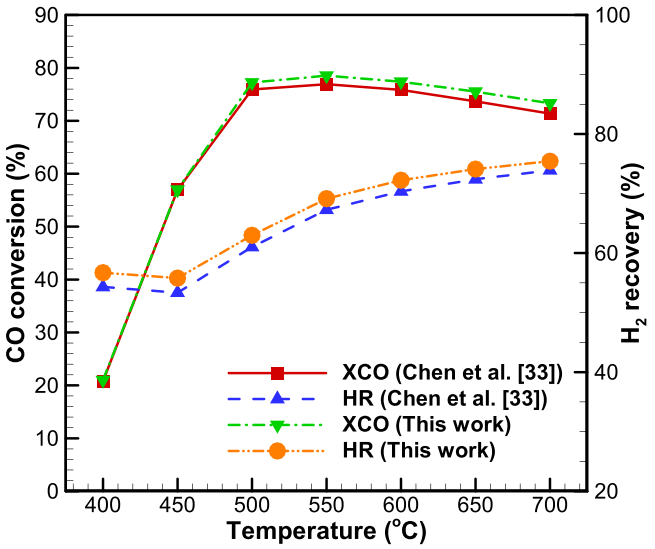


Fig. 4. Validation of the simulation model with the previous study.

where D_{ij} and τ are the binary gas diffusivity in free space and tortuosity of the catalyst pellet, respectively. The binary gas diffusivity is evaluated by the Chapman-Enskog equation [31,32].

In the porous energy equation, q_c is the energy source due to the chemical reaction and k_{eff} is the effective thermal conductivity of the catalyst bed and is defined as:

$$k_{eff} = \theta k + (1-\theta)k_{cat} \tag{8}$$

where k_{cat} is the thermal conductivity of the catalyst particle, k is the gas mixture thermal conductivity.

The boundary conditions in the computational domain are made up of (1) the inlet, (2) the centerline, (3) the reactor wall, and (4) the outlet, as shown in Fig. 1**c**. Details of the boundary conditions are shown in Table 2.

Numerical method and grid system

The computational model was built with the commercial software COMSOL Multiphysics 5.4. Fig. 2**a** is the grid distribution of the system for a single tube. Three different mesh numbers of 4, 096 (A), 6, 072 (B), and 10, 563 (C) were examined for the single tube system to test the requirement of grid independence. Due to the significant phenomena of mass transfer, heat transfer, and fluid mechanics around the catalytic membrane tubes and fluid zone, the catalytic layers were built with orthogonal grids. Fig. 2**b** shows a difference in the pressure coefficient (C_p) around the tube between the grid systems A and B. In contrast, the distributions almost overlap between the grid systems B and C. Fig. 3**a** shows the grid distribution of the four-tube system. Meanwhile, the C_p distributions along the surfaces of the first and fourth tubes under the three different grid systems are compared in Fig. 3**b** and Fig. 3**c**, respectively. The mesh structure was the same as the single tube (Fig. 2**a**). The results also reveal that grid system B is suitable for numerical simulation and is thus adopted. The developed numerical method is verified using grid system B and shown in Fig. 4. The profiles demonstrate that the simulation results match a previous study's data [33] very well at S/C = 2 and Re = 100, and the error for each concentration is below 5% between the simulations and the experiments. Accordingly, it validates the numerical method.

Chemical reaction

The conditions of the high-temperature WGSR were built according to a past study [33]. According to the study of Adams and Barton [34], the reaction rate, which used Fe-Cr oxide catalyst, was utilized and written as

$$r_{WGSR} = r_{forward} - r_{reverse} = \rho_{cat}(1-\theta)F_{press}\left(1.69 \times \frac{10^7 mol}{g{\cdot}h}\right)\times \exp\left(-\frac{88,000\ J/mol}{RT}\right)x_{CO}^{0.9}x_{H_2O}^{0.31}x_{CO_2}^{-0.156}x_{H_2}^{-0.05}(1-\beta) \tag{9}$$

where

$$\beta = \frac{x_{CO_2}x_{H_2}}{x_{CO}x_{H_2O}K_{eq}} \tag{10}$$

In Eq. (9), ρ_{cat} is the catalyst density and K_{eq} is the equilibrium constant, which was used for the reverse reaction rate, expressed as

$$K_{eq} = \exp\left(\frac{4577.8}{T} - 4.33\right) \tag{11}$$

A pressure scale-up factor (F_{press}) to account for the high-pressure effect on the reaction rate is expressed as [34]

$$F_{press} = P_r^{0.5-\frac{P_r}{250}} \tag{12}$$

where P_r is the pressure on the retentate side. For WGSR, the production rate of each species and energy source can be expressed as:

$$r_{CO} = r_{H_2O} = -r_{WGS}, r_{H_2} = r_{CO_2} = r_{WGS} \text{ and } q_c = r_{WGS}\Delta H_{WGS} \tag{13}$$

where ΔH_{WGS} is the heat of reaction, built by the temperature function [33], of reaction, expressed as

$$\Delta H_{WGS} = -4.912 \times 10^4 + \left(C_{pCO_2} + C_{pH_2} - C_{pH_2O} - C_{pCO}\right) \times (T - 298) \tag{14}$$

Table 3
Operating conditions and properties of membrane and catalyst.

Condition or property	Value
Temperature (°C)	500
Pressure (atm)	10
Compositions (%) [37]	H_2: 34.69 CO: 47.96 CO_2: 17.35
Re (-)	1, 5, 10, 20
S/C ratio (-)	1, 2, 4, 6, 8
Permeance ($mol{\cdot}m^{-2}{\cdot}s^{-1}{\cdot}Pa^{-0.5}$) [35]	1.62×10^{-3}
Membrane thickness (μm) [35]	3.2
Selectivity to CO, CO_2, and H_2O	0
Activation energy (kJ mol^{-1}) [35]	17.3
Selectivity to H_2	Infinite
Porosity (-) [36]	0.4
Catalyst density (kg m^{-3}) [34]	2476
Catalyst diameter (m) [36]	2×10^{-3}

Membrane properties and operating conditions

A Pd-Cu-Au ternary membrane was used to separate H_2 from the gas mixture [35]. For high-temperature applications, the Pd-alloy membrane with a permeance of $1.62 \times 10^{-2}\ mol{\cdot}m^{-2}{\cdot}s^{-1}{\cdot}Pa^{-0.5}$ was studied. In this study, the inlet temperature was maintained at 500 °C under the optimal range for CO conversion [33] to maximize the membrane's and the catalyst's performance. In the systems, attention was paid to the physical and chemical phenomena around the membrane tubes, so the Reynolds number ($Re = 2\rho\overline{V}R_o/\mu$) of the feed gas in terms of the membrane tube was given where ρ is the density, $2R_o$ is the diameter of the membrane tube, μ is the viscosity, and $\overline{V}$ is the mean flow velocity at the entrance of the duct. The operating conditions and the properties of the membrane [35] and the catalyst [34,36] are tabulated in Table 3. The feed gas consisted of three species: H_2 (34.69 vol%), CO_2 (17.35 vol

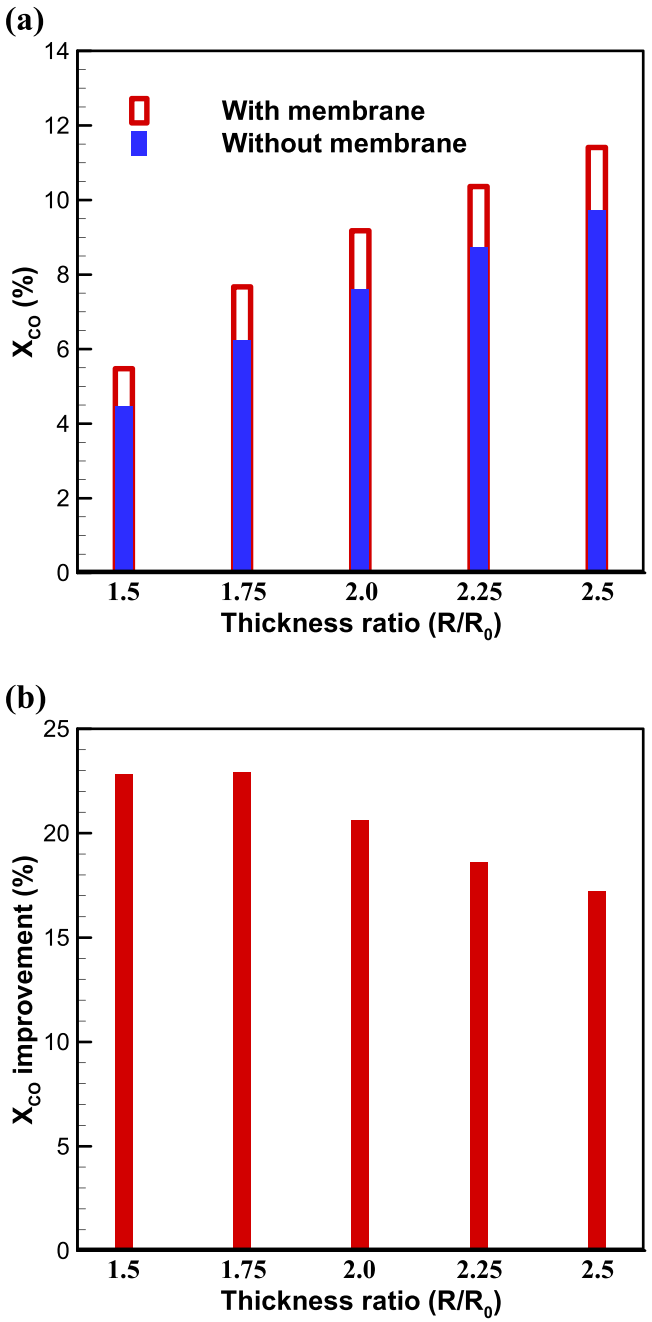


Fig. 5. (a) Comparison of CO conversion with and without membrane and (b) CO conversion improvement in different thickness ratios at $Re = 5$ and the S/C ratio = 4.

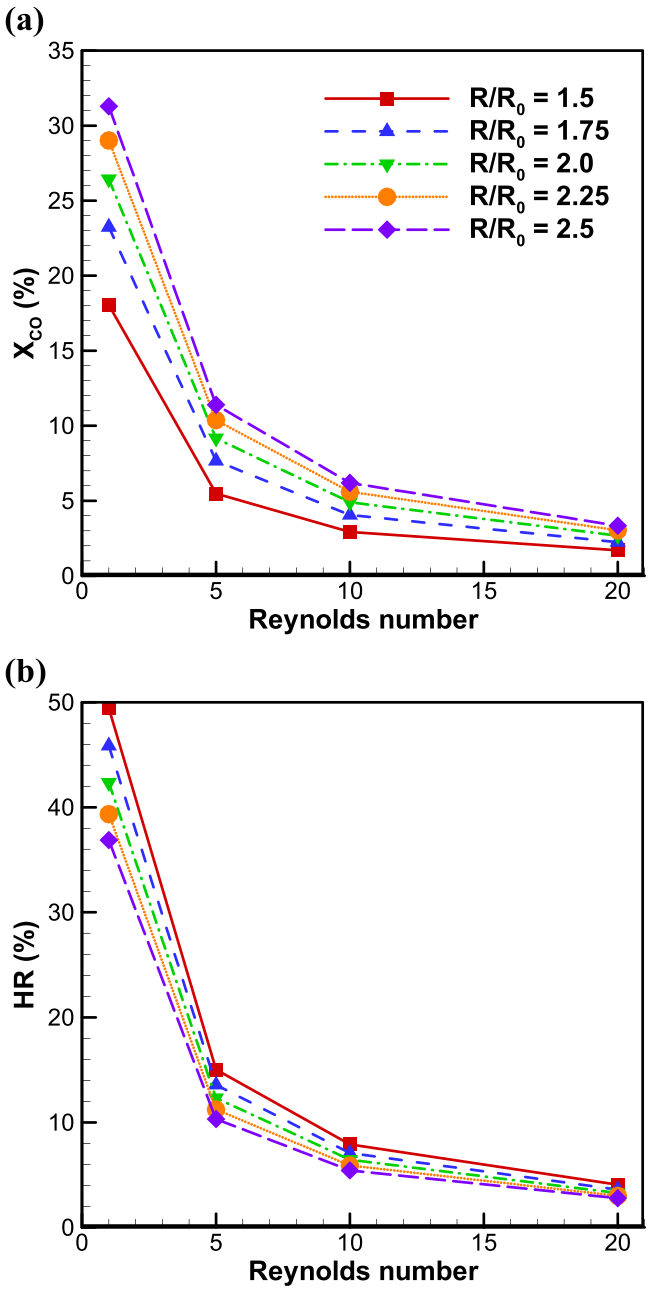


Fig. 6. Distributions of (a) CO conversion and (b) HR at different thickness ratios for the single tube system. (S/C ratio = 4).

%), and CO (47.96 vol%) from coal pure-oxygen gasification [37]. The S/C ratios of 1, 2, 4, 6, and 8 were used to investigate the optimal conditions for this reactor. The retentate side pressure was 10 atm to keep the high-pressure environment for WGSR and hydrogen permeation.

Results and discussion

Influence of membrane

The performance of the Pd membrane is important for the reactors to increase H_2 production and hydrogen recovery (HR). This study uses the CO conversion, HR, and CO conversion improvements to investigate the membrane performance. They are defined as follows:

$$\text{CO conversion } (\%) = \frac{CO_{inlet} - CO_{outlet}}{CO_{inlet}} \times 100 \tag{15}$$

$$\text{HR } (\%) = \frac{H_{2,permeation}}{H_{2,permeation} + H_{2,outlet}} \times 100 \tag{16}$$

$$\text{CO conversion improvement } (\%) = \frac{CO\ conv._{\text{w/ membrane}} - CO\ conv._{\text{w/o membrane}}}{CO\ conv._{\text{w/o membrane}}} \times 100 \tag{17}$$

For the different thicknesses of the catalytic layer, the R/R_0 value from 1.5 to 2.5 is regarded. The influence of the Pd membrane is measured by comparing the CO conversion at $Re = 5$ and S/C ratio = 4 with/without the membrane in the catalytic tube. The performance of CO conversion significantly increases when the membrane is coated at the tube surface, as shown in Fig. 5**a**. The highest CO conversion is

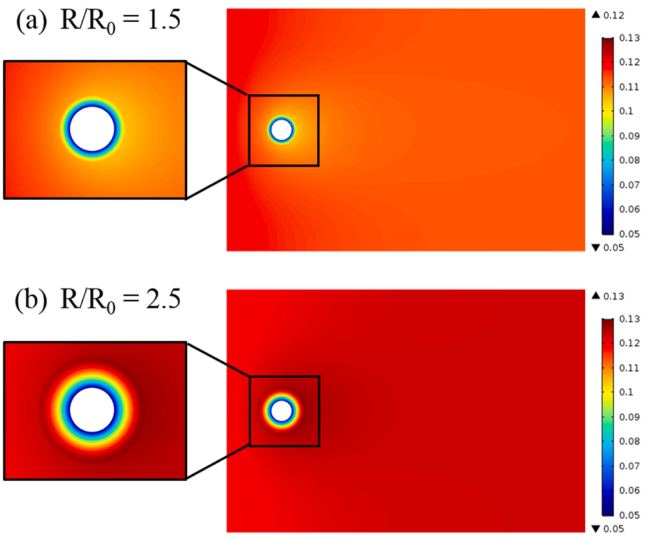


Fig. 7. The contours of H_2 concentration distribution with (a) $R/R_0 = 1.5$ and (b) $R/R_0 = 2.5$ at $Re = 5$ and S/C ratio = 4.

11.41% by one tube at $R/R_0 = 2.5$. One catalytic tube's CO conversion is 9.73% at the same thickness ratio. The observed outcome can be attributed to applying Le Chatelier's principle, wherein the removal of hydrogen facilitates the WGSR, leading to an increase in CO conversion rate [8]. Other literature sources have confirmed that membrane tubes enhance the conversion rate. For example, Sloria et al. [38] investigated the membrane reactor with low-temperature WGSR, where adding a membrane increased CO conversion by about 5%. Li et al. [27] observed the reactor overcame the thermodynamic equilibrium limitations by H_2 removal (collect) with a Pd membrane. To identify the efficiency of the catalytic layer, the CO conversion improvement is defined by the difference in CO conversion in the case with and without membrane divided by the CO conversion without the membrane. The profile of CO conversion improvement is shown in Fig. 5**b**. The greatest CO conversion improvement is around 22.9% at $R/R_0 = 1.75$, and the lowest improvement is 17.2% at $R/R_0 = 2.5$. It represents that the membrane tube can intensify the CO conversion efficiently when the thickness ratio is 1.75. However, as the catalytic layer thickens, the efficiency decreases due to slightly increasing CO conversion. Dong et al. [39] also obtained results for CO conversion improvement by membrane addition. The results proved that the membranes improved WGSR by recovering H_2. Uemiya et al. [40] also observed a similar phenomenon to improve the reactor when the membrane was used. Zhang et al. [41] illustrated that CO conversion was improved, which could be attributed to the deviation of the reaction system from the equilibrium state.

Influences of catalytic layer thickness and reynolds number

The influences of varying catalytic layer thickness and Reynolds number on CO conversion and HR at S/C = 4 of the single tube system are shown in Fig. 6 for comparison. There is a significant increase in CO conversion as R/R_0 increases from 1.5 to 1.75, as shown in Fig. 6**a.** It represents a larger efficiency improvement. At Re=1, the CO conversion is 18.05% and 23.24% at R/R_0=1.5 and 1.75, respectively. At the same Reynolds number, the highest CO conversion is 31.3% at R/R_0=2.5. However, with increasing R/R_0, its impact on CO conversion improvement withers. This is because the volume of the annular catalyst approaches the critical value for CO conversion.

In contrast, the impact of the Reynolds number on CO conversion is significantly greater than the influence of the catalytic layer thickness. A lower Reynolds number leads to a higher contact time between the catalyst and reactants, thereby promoting CO conversion. Yin et al. [42] studied space speed's influence and concluded that low velocity could

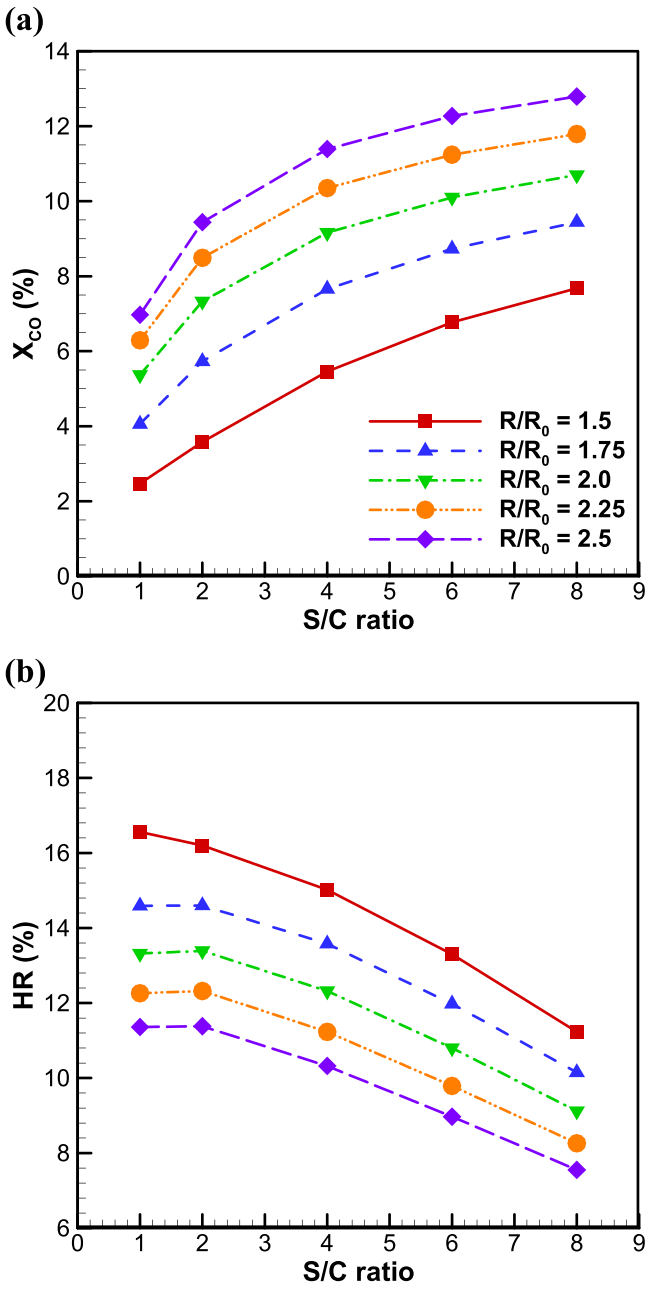


Fig. 8. Distributions of (a) CO conversion and (b) HR at different S/C ratios for the single tube system. (Re = 5).

increase CO conversion but needed to avoid the concentration polarization phenomenon. Chu et al. [43] suggested the contact time for the catalyst was a way to control the CO concentration at the retentate side. Increasing the contact time in the catalyst could decrease the CO concentration efficiently. At Re=5, the CO conversion is 11.4% when the R/R_0 is 2.5. The results show that the performance reduces by almost 20% compared to the result of Re=1.

On the other hand, HR decreases as the catalytic layer thickens and the Reynolds number increases, as shown in Fig. 6**b**. The maximum HR is almost 50% at R/R_0=1.5 and Re = 1. When the Reynolds number increases to 5, HR is lowered to 15% at R/R_0=1.5. This means the Reynolds number is a very influential parameter for reaction and permeation. The relationship between CO conversion and the catalytic layer thickness follows a trend opposite to that of HR and the catalytic layer thickness. The opposite trend can be attributed to the excess H_2 produced by thickening the catalytic layer. The excess H_2 also caused the

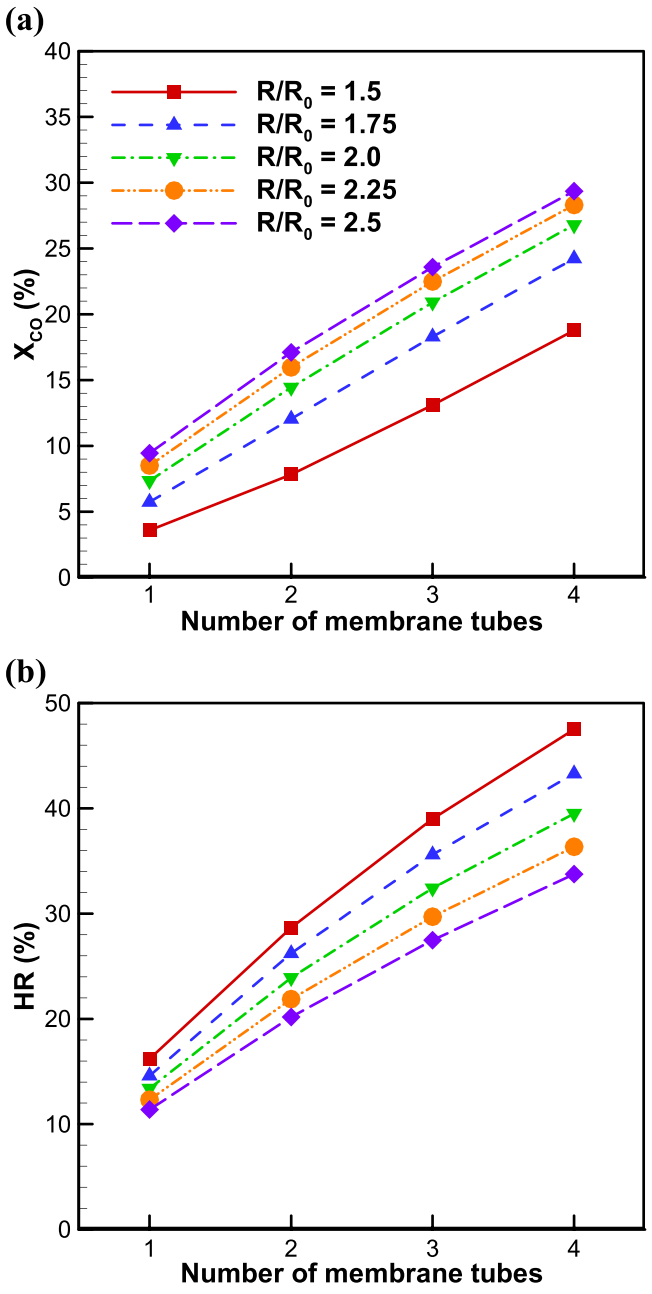


Fig. 9. Distributions of (a) CO conversion and (b) HR at different S/C ratios for the single tube system. (*Re* = 5 and S/C ratio = 2).

membrane to be unable to receive all of it. Fig. 7 shows the H_2 concentration contours at *Re*=5 and S/C = 4 under R/R_0=1.5 and 2.5. The thicker catalytic layer (R/R_0=2.5, Fig 0.7b) produces more H_2 than the thinner one (R/R_0=1.5, Fig 0.7a) when the feed gas contacts the catalytic layer, revealing the better performance to produce H_2 from the thick catalytic layer tube system.

Influence of S/C ratio

Since the inlet speed is too slow at *Re*=1, *Re*=5 is chosen in this study. Fig. 8 shows the CO conversion and HR under different S/C ratios and R/R_0 values at *Re* = 5. The CO conversion is noticeably enhanced by increasing the S/C ratio and R/R_0, as shown in Fig. 8**a**. As the S/C ratio increases, the CO conversion also increases. It can be explained by the equilibrium state being pushed toward the right side of the WGSR when more steam is added to the reaction system. Sanz et al. [44] investigated the S/C ratio for low-temperature WGSR and had a similar trend in a membrane reactor. They found that increasing the S/C ratio led to an enhanced CO conversion. The higher steam content in the reaction system promotes the formation of additional water molecules, which react with CO to produce CO_2 and H_2. Therefore, in this study, the increase in CO conversion with increasing S/C ratio can be attributed to the favorable shift in the WGSR equilibrium resulting from the higher steam content. In different S/C ratios, the highest CO conversion is 12.8% for the one-tube system at R/R_0=2.5 and S/C = 8.

For all the R/R_0 ratios, the best HR performance develops at S/C = 2, except for R/R_0=1.5, as shown in Fig. 8**b**. For R/R_0=1.75, 2.0, 2.25, and 2.5, a slight increase in HR is exhibited as S/C increases from 1 to 2. This is mainly attributed to the increased H_2 partial pressure, leading to a slight increase in hydrogen permeation. However, the CO conversion is relatively insensitive to S/C at R/R_0 = 1.5, so it gives an exception in the increasing trend of HR. The highest HR is 16.56% at R/R_0=1.5 and S/C

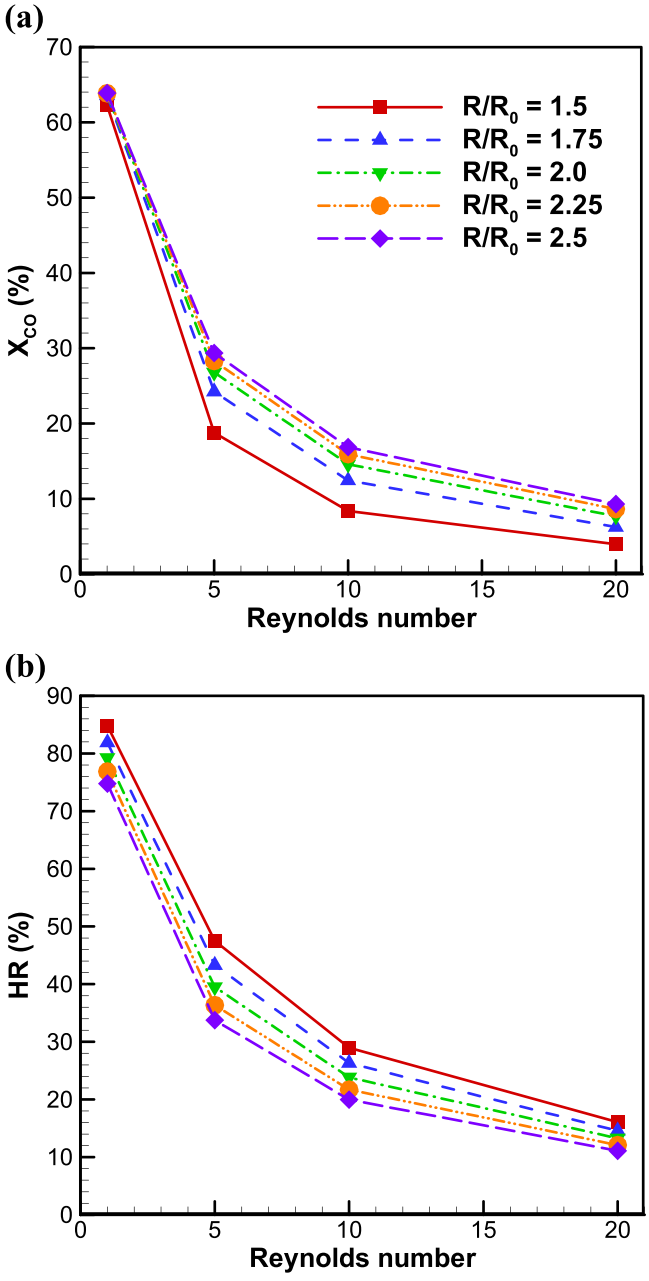


Fig. 10. Distributions of (a) CO conversion and (b) HR at different S/C ratios for the four tandem tubes system. (S/C ratio = 2).

= 1. When the S/C ratio is beyond 2, its increase lowers HR. Two reasons should be pointed out. First, adding extra steam at a high S/C ratio causes H_2 dilution. Secondly, previous research by Augustine et al. [45] increasing the S/C ratio leads to excessive H_2 production, which negatively impacts HR. Additionally, we considered the higher temperature associated with higher S/C ratios, which can adversely affect the system's stability and long-term operation.

Chein and Yu's study [46] provided valuable insights into the relationship between the S/C ratio and carbon accumulation. They proposed that a higher S/C ratio resulted in a carbon-free WGSR due to the reaction occurring at higher temperatures. The elevated temperatures facilitated the cracking and gasification of carbon species, preventing their accumulation on the catalyst surface. This finding aligns with our objective of increasing CO conversion and reducing carbon accumulation. By selecting an S/C ratio of 2, we aim to maintain a sufficiently high temperature for carbon-free WGSR while avoiding excessive H_2 production and HR degradation. This approach allows us to balance the system's CO conversion performance, carbon management, and long-term stability.

Influence of tandem tube numbers

Fig. 9 compares CO conversion and HR under different numbers of tubes with the tandem configuration at *Re*=5 and S/C = 2. It is unsurprising that CO conversion and HR increase as more tubes are added. The highest CO conversion is 29.36% for four tubes at R/R_0=2.5. On the other hand, the highest value of HR is 47.50% for the four-tube system at R/R_0=1.5. The performance almost linearly increases when the number of tubes increases. Thus, increasing the number of tubes can potentially improve performance. However, the difference in CO conversion gradually decreases as the R/R_0 ratio goes up. This implies a limit to the increase in performance if the catalytic layer increases continuously. The results also reflect that the number of membrane tubes has a more significant influence than the catalyst layer thickness in this reactor. As shown in Fig. 9**a**, the CO conversion gap between R/R_0=1.5 and 1.75 is lifted gradually from a single tube to four tubes. The results suggested that increasing the number of tubes and thickening the catalytic layers can improve the reactor system's performance. Although the best performance of HR is 47.5% at R/R_0 = 1.5, the CO conversion performance is not outstanding. At the R/R_0 = 2.5, HR is 33.75% for four tubes. Considering the performance, the R/R_0 ratio higher than 1.5 can be applied.

Influence of Reynolds number for four tandem tubes

This study computes the effects of different thickness ratios and Reynolds numbers on four tandem membrane tube systems to improve this reactor's CO conversion and HR performance. The CO conversion values of all R/R_0 are close to 64% at the lowest Reynolds number (i.e., *Re*=1), as shown in Fig. 10**a**. That can be considered that the CO conversion for the four-tube system has a limit at a low Reynolds number. According to a past study [25], the low-velocity zone for the four tubes in tandem further extended downstream after the first one, implying that the role played by the wake flows in the physical phenomena is more pronounced, which induces a wall boundary layer effect [47]. That causes a higher contact time and promotes WGSR [48], leading to the CO conversion being close to 64%. The CO conversion decreases as the Reynolds number increases, and HR has the same trend. An HR value up to 84.80% occurs at *Re*=1 and R/R_0=1.5, as shown in Fig. 10**b**. When the Reynolds number increases to 5, CO conversion decreases to almost 30% at R/R_0 = 2.5 and the HR decreases to 47.5% at R/R_0=1.5. The results suggest that the influence of the Reynolds number on the CO conversion is more significant than the catalyst thickness in the multi-membrane system.

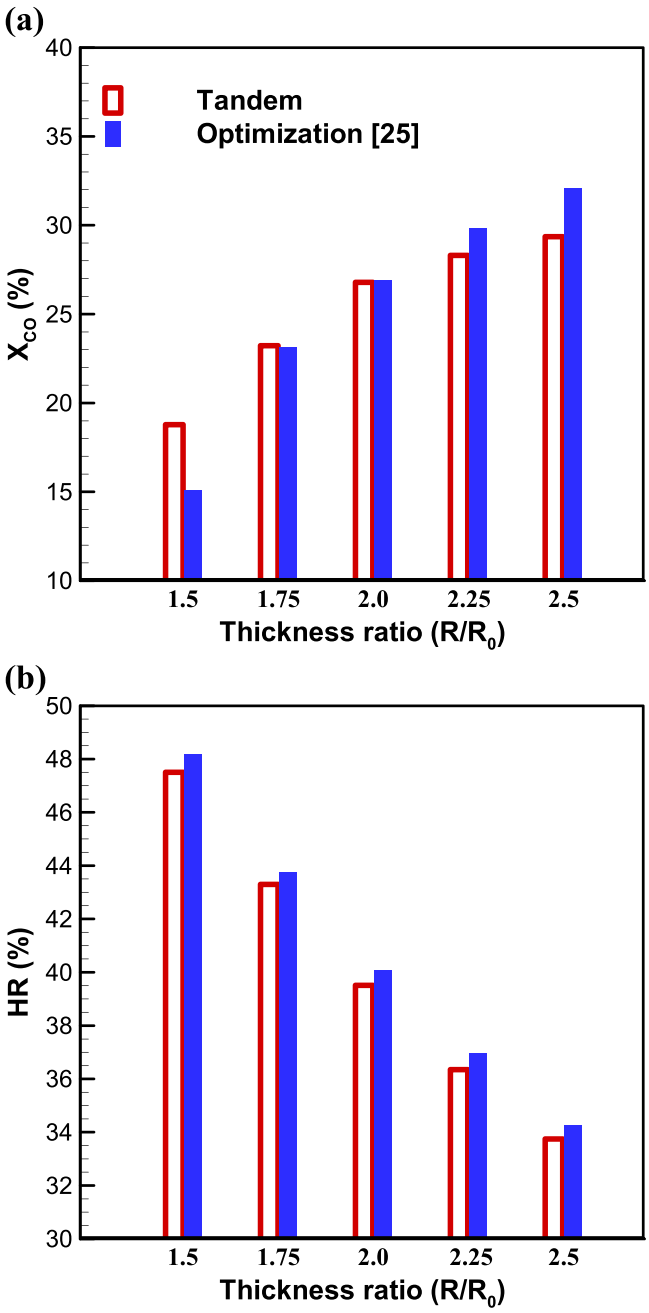


Fig. 11. Comparison between tandem and optimized configurations with (a) CO conversion and (b) HR in four tubes system.

Optimized configuration

In a previous study, Chen et al. [25] used evolutionary computation to design the configuration of a multiple membrane tube system. According to their research, this study used the same optimization configuration compared to the tandem configuration. As shown in Fig. 11**a**, there are obvious gaps at R/R_0=1.5, 2.25, and 2.5. The CO conversion in the tandem configuration at R/R_0 = 1.5 is 18.78% and higher than the optimization case of 15%. Conversely, the CO conversion values at R/R_0=2.25 and 2.5 in the optimized configuration are 29.8% and 32.1%, respectively. The CO conversion values are raised by 2% and 3%, respectively, compared to the tandem configuration. Fig. 11**b** suggests that the optimized design has better performance in recovering H_2 than the tandem configuration. This can be explained by the weaker influence of the catalytic thickness, and the optimized

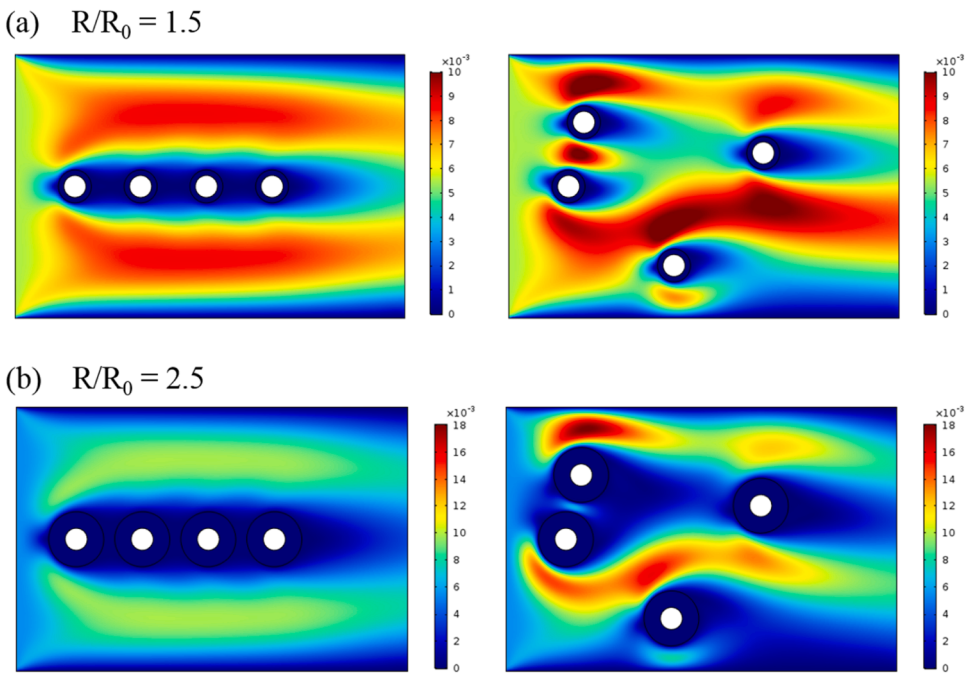


Fig. 12. The contours of velocity with (a) $R/R_0 = 1.5$ and (b) $R/R_0 = 2.5$ in tandem and optimization at $Re = 5$ and S/C ratio = 2.

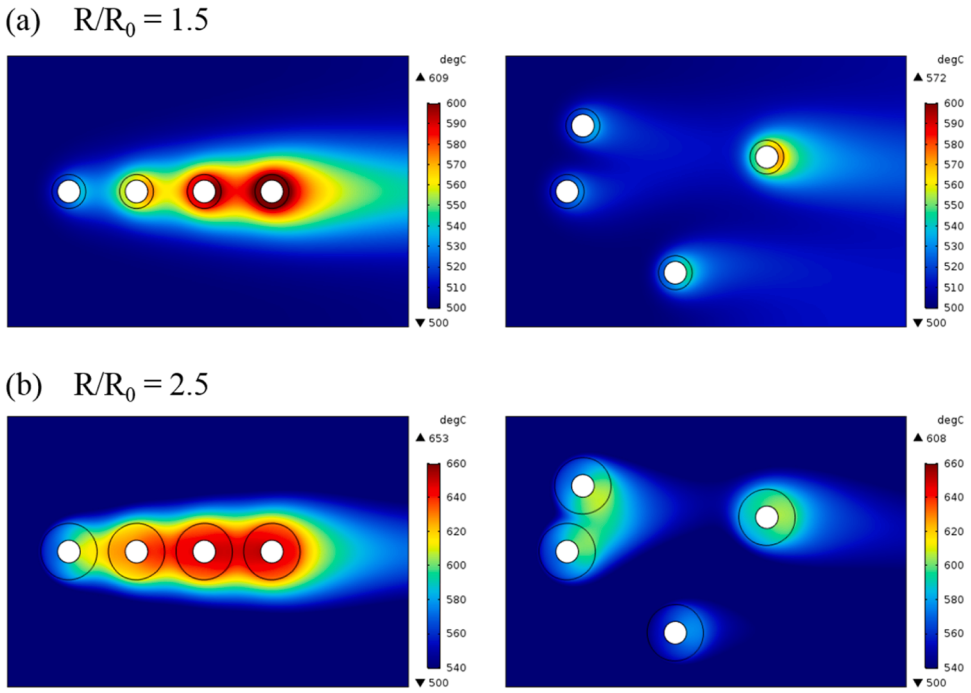


Fig. 13. The contours of temperature distribution with (a) $R/R_0 = 1.5$ and (b) $R/R_0 = 2.5$ in tandem and optimization at $Re = 5$ and S/C ratio = 2.

configuration can minimize the concentration polarization simultaneously [25]. The HR is 48.2% at R/R_0=1.5 and 34.3% at R/R_0=2.5.

According to the velocity contours shown in Fig. 12**a**, the tandem configuration causes a low-speed zone by the wake flow, increasing the temperature at Re=5 and R/R_0=1.5, as shown in Fig. 13**a**. The optimized configuration avoids the effect of wake flow, and the temperature increase is less than tandem one. The higher temperature can push the reaction forward to get better CO conversion performance. It could be considered that the low speed is a positive influence when the catalytic layer is thin. The low-speed zone is between each tube at R/R_0=2.5 in the tandem configuration, as shown in Fig. 12**b**. And the average temperature is higher than the optimized one, as shown in Fig. 13**b**. The results show that the higher temperature also appears in the tandem configuration, but the performance is not better than the optimized one. It means that temperature is not a key factor in increasing CO conversion when the catalyst layer thickness increases.

Fig. 14**a** shows the reaction rate contours at R/R_0=1.5. It displays that the reaction rate in the tandem configuration is higher than in the optimized configuration. The reaction rate grows downstream in the tandem configuration due to the wake flow effect. In contrast, the

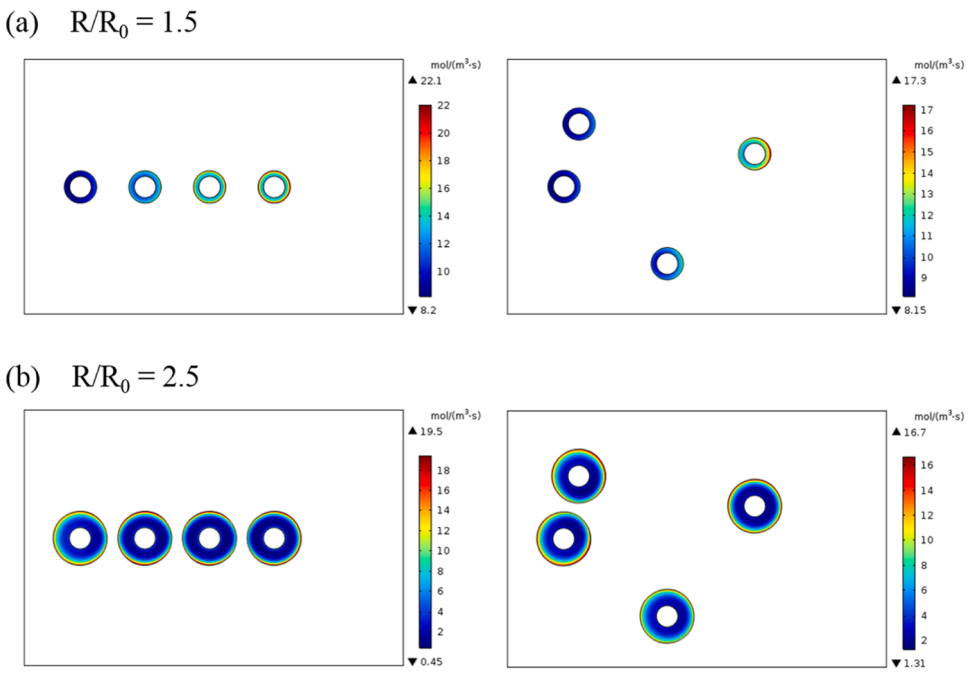


Fig. 14. The contours of reaction rate with (a) $R/R_0 = 1.5$ and (b) $R/R_0 = 2.5$ in tandem and optimization at $Re = 5$ and S/C ratio = 2.

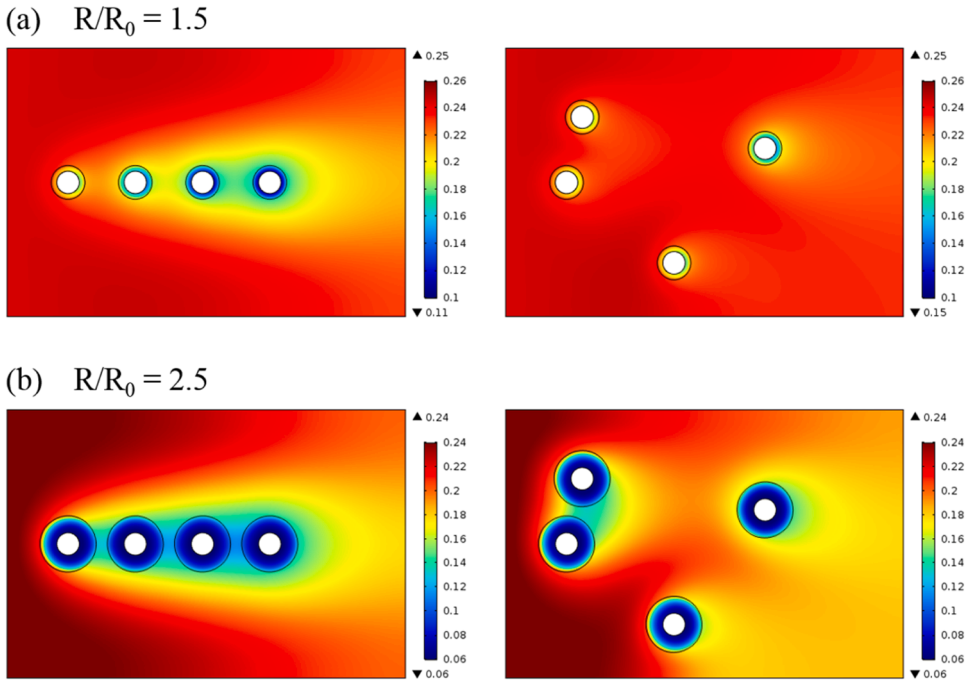


Fig. 15. The contours of CO concentration distributions with (a) $R/R_0 = 1.5$ and (b) $R/R_0 = 2.5$ in tandem and optimization at $Re = 5$ and S/C ratio = 2.

reaction rate around the final tube in the optimized configuration is much higher than the first three tubes. On the other hand, the reaction in the tandem configuration is concentrated at the first tube for R/R_0=2.5, as shown in Fig. 14**b**. It is due to the negative effect of the low speed. Relatively, the optimized configuration avoids this negative effect. The negative effect causes CO to be consumed when it comes in contact with the first tube, as shown in Fig. 14**b**. The remaining CO bypasses the CMRs. As seen in Fig. 15**a**, the CO concentration decreases in the tandem configuration more clearly than in the optimized configuration at R/R_0=1.5 via the CMRs. Fig. 15**b** shows that the CO concentration becomes the limiting factor as it decreases intensely and affects the performance downstream in the tandem configuration. This phenomenon is not obvious in the optimized configuration. According to the results, the low speed has a positive effect on the thin catalytic layer and becomes a negative effect on the thick catalytic layer.

Conclusions

This study used computational fluid dynamics (CFD) to simulate the water gas shift reaction at high temperatures in a catalytic membrane reactor with a Fe-Cr catalyst. The effect of adding a Pd membrane was investigated, as was the catalytic layer thickness ratio, the Reynolds

number, and the steam-to-CO ratio. These parameter adjustments were evaluated by looking at the CO conversion and hydrogen recovery. The important

- Adding a Pd membrane can efficiently improve CO conversion in the WGSR reactor system. The performance of a system with a Pd membrane increases to 22.9% when the thickness ratio of the catalytic layer (R/R_0) is 1.75.
- The impacts of Reynolds number and S/C ratio on a single tube configuration's reaction and H_2 recovery (HR) are studied. The CO conversion and HR decline when the Reynolds number increases for the CMR because of insufficient contact time with the catalyst during the WGSR. A larger catalytic layer has better CO conversion efficiency, up to 11.4%, but an increase of H_2 from the conversion causes H_2 dilution due to the limited permeation of the membrane. The profiles of the S/C ratio suggest that elevated steam (higher S/C ratio) causes HR to drop while the CO conversion increases.
- A tandem of one to four tubes is compared in multiple trials. The linear results show that increasing the number of membrane tubes improves the system's performance. The Reynolds number influences CO conversion and hydrogen recovery more than the catalyst layer thickness in the four-membrane tube system. A lower Reynolds number provides more contact time for WGSR, leading to slight differences in CO conversion.
- The optimized configuration performs better for HR at every thickness ratio than the tandem configuration. This revealed that the optimized configuration improves CO conversion performance regardless of the catalytic layer thickness. Choosing the appropriate catalytic layer thickness and Reynolds number can improve the reactor based on the optimized configuration.

The results all indicate that optimizing operating parameters for a catalytic membrane reactor is vital for efficiently using the water gas shift reaction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgments

The authors would like to acknowledge the financial support of the National Science and Technology Council, Taiwan, R.O.C., under the grant numbers NSTC 112-2221-E-006-111-MY3 for this research. This research is also, in part, supported by the Ministry of Education, Taiwan, R.O.C., Headquarters of University Advancement to the National Cheng Kung University (NCKU).

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